# Rehabilitation Center Visit — Plan

Kennedy Krieger Institute, Baltimore MD | November 10, 2026 | Daniel Dobin, Clinical Assistant II, Neurobehavioral Unit (Inpatient)

## 1. What this visit has to produce

Four things are required downstream, so the visit is not finished until each has been captured. The first two are named explicitly in the project description.

- **Needs with a priority ranking.** The Preliminary Design Review requires a Kano diagram, and the description states that the clinical or client interview should guide the needs prioritization. A list of needs without a ranking does not satisfy this.
- **Reaction to solution directions.** The concept screening must include feedback from clinical contacts on the chosen concept. Whatever is said today about the three broad directions becomes the clinical input to the Pugh matrix later.
- **Benchmarking numbers.** Cost, service life, setup time, staff count per task, repair frequency. These populate the state-of-the-art section and the PDS.
- **Observation notes.** Written up the same day, while detail is still recoverable.

## 2. Running order (45—60 minutes)

**Opening, 5 min.** Introduce yourself as a Hopkins engineering graduate student in a rehabilitation engineering course. State the project in two sentences. **Ask permission before taking any notes, sketches or photographs** — this is a behavioral inpatient unit, patient privacy governs, and the answer on photography is likely no. Follow staff direction on where you may go without further discussion.

**Tour and observation, 20 min.** The project description asks you to observe a patient in evaluation and treatment with members of the rehab team. Treat it as the Module 8 exercise: watch for the task sequence, not the diagnosis.

- Note the **step sequence** of whatever activity you see, in order. You are building a task analysis, so capture the steps even if the activity is unrelated to your project.
- Mark the steps that are **difficult**, and why — hesitation, repeated effort, staff prompting, a pause before committing.
- Where possible, **count and time**: attempts, seconds for the hard step, seconds for an ordinary step beside it. The ratio is the useful number.
- Note **where staff assistance concentrates**. That is where the design opportunity is.
- Ask whether the **therapy garden** can be included. Its gravel, grass and cobblestone surfaces are the closest thing on site to your operating environment.

**Hands on the equipment, 10 min.** The description asks you to experience assistive technology close to your project. Ask to handle anything that attaches to a chair.

- For each: **weight, how it attaches, how long setup takes, how it releases**, and whether you could do it one-handed.
- Note what it is **made of and how it is cleaned**. On this unit, cleanability is a requirement rather than a preference.

**Interview, 20 min.** Work from the submitted question list. If time collapses, the six below are the ones to protect.

**Close, 5 min.** Ask for the referral to a PT, OT or seating and mobility specialist (question 26). Ask whether you may follow up by email with anything you missed, and get an address. Thank him, and say what you will send back.

## 3. The six questions to protect if time runs short

- **Q14 — hard safety requirements for anything brought onto the unit**: cleanability, ligature risk, pinch points, parts that detach or can be thrown. Nobody else you meet will answer this as well, and these are real design constraints.
- **Q7 — rank durability, setup speed, cleanability, safety with limited supervision, holding interest, and cost.** Push for essential / important / nice to have. This is the Kano input and it is required.
- **Q4 — how devices fail**: breakage, batteries, lost parts, too fiddly, or patients simply disengage. Abandonment is more informative than adoption.

- **Q16 — the three solution directions.** Carry the tool and its reaction force on the chair; carry the force into the ground separately; or change the task so a large force never arises. Record the reasoning, not just the preference.
- **Q12 — staff count per activity, and what would reduce it to one.** A quantified metric you can design against.
- **Q26 — the referral.** Everything about stability, propulsion and funding sits with a clinician you have not met yet.

## 4. Be upfront about two things

Both will come up, and both are better volunteered than discovered.

- **You have not converged on a design.** The project description frames this visit as gathering input on the problem *and* the proposed solution, but concept generation for this project has not happened yet. Say that the three directions are deliberately broad and that you would rather leave with a fourth you had not considered.
- **Your population is adults; this unit is pediatric and behavioral.** Rather than work around it, use question 25: what happens at the age-21 transition, how much planning covers independent living at home, and where patients source equipment after they age out. That turns the mismatch into the most interesting question you ask.

## 5. Within thirty minutes of leaving

Before anything else, sit down and write. Specifically:

- The **task sequence** you observed, in order, with the difficult steps marked — this is the raw material for a task analysis diagram.
- Any **numbers** you heard, with units and the question they answered.
- The **needs ranking**, exactly as he ordered it, including anything he refused to rank.
- **Direct quotes.** Two or three sentences in his own words are worth more in the PDR than a page of paraphrase.
- **What surprised you.** The thing you did not expect is usually the finding.
- Anything you **could not get**, and who you would need to ask.